

MovieIQ

Discover • Explore • Watch

FILM SUCCESS INTELLIGENCE

DATA ANALYTICS • PREDICTIVE MODELING • INTERACTIVE INSIGHTS

Uncover the data behind box office success. Analyze trends, explore key factors, and predict which movies are destined to hit big.

- EXPLORATORY DATA ANALYSIS
- VISUAL INSIGHTS
- MACHINE LEARNING PREDICTIONS
- FEATURE IMPORTANCE
- STATISTICAL TESTING



PREPARED BY

AYUSH DASH

DATE
19th August ,2026

DATASET
movies.csv

TOOLS & TECHNOLOGIES
Python • Streamlit
Pandas • Scikit-learn • Plotly

DISCOVER. EXPLORE. WATCH. PREDICT.

TURNING DATA INTO BLOCKBUSTERS!

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Executive Summary

MovieIQ — Film Success Intelligence is an interactive business intelligence and predictive analytics platform developed to analyse historical movie performance and support evidence-based decision-making in the film industry. The project integrates data preparation, exploratory data analysis, financial analysis, genre intelligence, audience analysis, statistical testing, machine learning, and interactive dashboarding using Python, Pandas, NumPy, Plotly, SciPy, Scikit-learn, and Streamlit.

The analytical dataset contains 2,000 movies and 19 fields, with 2,000 valid financial records available for financial and success analysis. Movie success is defined using the analytical rule Revenue > Budget. Under this definition, 80.7% of the analysed movies are classified as successful. The portfolio generated approximately $356.11 billion in total revenue against $203.84 billion in total budget, resulting in $152.27 billion in revenue-minus-budget profit and an average ROI of 75.4%.

A key finding is the strong positive relationship between budget and revenue, with Pearson correlation of $r = 0.760$ and $p \approx 0.000$, indicating a statistically significant linear association. The independent t-test identified a statistically significant difference in popularity between successful and unsuccessful movies ($p = 0.03968$). However, the Chi-Square test found no statistically significant association between primary genre and the defined success outcome ($p = 0.99457$).

MovieIQ also implements a Random Forest classifier using budget, popularity, runtime, vote average, and primary genre. The displayed model evaluation reports 65.0% accuracy, 80.2% precision, 75.2% recall, and 77.6% F1-score, while the ROC-AUC of 0.473 indicates limited discrimination capability. Therefore, the model should be considered a decision-support and screening tool rather than a guaranteed forecasting system.

Overall, MovieIQ demonstrates how historical movie data can be transformed into financial, audience, statistical, and predictive intelligence for portfolio benchmarking, project screening, and informed decision-making.



Figure – KPI

Table 1. Executive Summary — Key Analytical Indicators

Indicator	Result	Interpretation
Movies Analysed	**2,000**	Historical movie sample
Total Revenue	**$356.11B**	Overall reported revenue
Total Budget	**$203.84B**	Combined production budget
Success Rate	**80.7%**	Revenue exceeded budget
Average ROI	**75.4%**	Average return relative to budget
Budget–Revenue Correlation	**r = 0.760**	Strong positive relationship
Popularity T-Test	**p = 0.03968**	Statistically significant difference
Genre Chi-Square	**p = 0.99457**	No significant association
Random Forest Accuracy	**65.0%**	Moderate classification performance
ROC-AUC	**0.473**	Limited discrimination capability

Chapter 1 — Business Context

1. Introduction

The film industry is characterized by substantial financial uncertainty, where production decisions are often made before the actual revenue, audience response, and commercial performance of a movie are known. Producers, studios, investors, and marketing teams must therefore evaluate multiple factors when assessing the potential performance of a film. Traditional decision-making based only on individual experience or isolated financial measures may not provide a complete understanding of the patterns present in historical movie data.

MovieIQ — Film Success Intelligence was developed to address this challenge through a structured data analytics and business intelligence approach. The project analyses historical movie information including budget, revenue, popularity, runtime, vote average, vote count, genre, and release-related attributes. These variables are examined to understand financial performance, audience behaviour, genre patterns, and factors associated with movie success.

The project defines analytical success as **Revenue > Budget**, creating a binary success indicator that can be used for comparative analysis and predictive modelling. MovieIQ combines exploratory data analysis with financial metrics such as profit and Return on Investment (ROI), allowing movies to be evaluated from both revenue and investment-efficiency perspectives. Statistical techniques, including Pearson correlation, independent t-test, and Chi-Square testing, are applied to determine whether observed relationships and group differences have statistical evidence.

In addition, a Random Forest classification model is implemented using selected historical movie characteristics to provide predictive screening support. The findings are presented through an interactive Streamlit dashboard, enabling users to explore portfolio-level performance as well as individual movie information. Overall, MovieIQ demonstrates how historical entertainment data can be transformed into structured analytical intelligence to support more informed and evidence-based film decision-making.

2. Business Problem

The film industry involves significant financial uncertainty because major investment decisions are frequently made before the actual commercial performance of a movie is known. Producers, studios, and investors must determine whether a proposed project has characteristics associated with stronger financial outcomes, while marketing teams need to understand how audience engagement relates to movie performance. Without systematic analysis, these decisions may rely heavily on intuition, historical experience, or limited financial indicators.

A major business challenge is therefore to identify measurable patterns that distinguish stronger-performing movies from weaker outcomes. MovieIQ addresses this challenge by

analysing historical movie characteristics such as production budget, revenue, popularity, runtime, vote average, vote count, and genre. The analysis examines whether financial investment is associated with revenue, whether different genres demonstrate different historical performance patterns, and whether audience indicators such as popularity and ratings are related to financial outcomes.

The project also addresses the challenge of defining and measuring movie success consistently. For analytical purposes, MovieIQ classifies a movie as successful when **Revenue > Budget**. This creates a binary outcome that can be used for group comparisons, statistical testing, and predictive modelling. However, this definition represents an analytical financial signal rather than complete accounting profitability because factors such as marketing expenditure, distribution costs, taxes, financing, and revenue-sharing arrangements are not included.

Another challenge is converting analytical findings into practical decision-support tools. MovieIQ therefore combines statistical evidence with a Random Forest classification model and an interactive Streamlit dashboard. The objective is not to eliminate uncertainty or guarantee box-office success, but to provide a structured framework through which historical patterns can be explored, projects can be benchmarked, and potential risks can be assessed using evidence rather than isolated assumptions.

3. Objectives

The primary objective of MovieIQ — Film Success Intelligence is to develop an analytical and interactive decision-support platform that uses historical movie data to identify financial, audience, genre, and predictive patterns associated with movie performance. The project aims to transform raw movie data into structured insights that can support evidence-based decision-making for producers, studios, investors, and other stakeholders.

The specific objectives of the project are:

1. **To prepare and validate historical movie data** by handling missing values, converting variables into appropriate analytical formats, validating financial records, and creating a reliable dataset for analysis.
2. **To define and measure movie success** using the analytical rule **Revenue > Budget**, enabling consistent comparison between successful and unsuccessful movies.
3. **To analyse financial performance** through total revenue, budget, profit, ROI, and success rate, with particular focus on the relationship between production budget and revenue.
4. **To examine genre performance** using measures such as movie count, average revenue, average budget, ROI, and success rate to identify historical genre-level patterns.
5. **To investigate audience intelligence** by analysing popularity, vote average, vote count, revenue, and success outcomes.
6. **To identify potential success drivers** through correlation analysis and comparative analysis of successful and unsuccessful movies.
7. **To apply statistical testing** using an independent t-test, Chi-Square test, and Pearson correlation to determine whether observed group differences and relationships have statistical evidence at a significance level of $\alpha = 0.05$.

8. **To develop and evaluate a Random Forest classification model** using selected movie characteristics to provide predictive screening support.
9. **To evaluate model performance** using accuracy, precision, recall, F1-score, ROC-AUC, confusion matrix, ROC curve, and feature importance.
10. **To develop an interactive Streamlit dashboard** that presents analytical findings, movie-level exploration, statistical evidence, and predictive outputs in an accessible business intelligence environment.

Overall, the project aims to convert historical movie data into measurable and actionable intelligence while maintaining clear boundaries between statistical association, prediction, and guaranteed commercial success.

4. Scope and Analytical Questions

The scope of MovieIQ — Film Success Intelligence covers the analysis of historical movie-level data to understand financial performance, audience behaviour, genre patterns, statistical relationships, and predictive indicators of movie success. The analysis focuses on variables available in the dataset, including budget, revenue, popularity, runtime, vote average, vote count, genre, and release-related information. Financial analysis is based on reported production budget and revenue, while movie success is defined as **Revenue > Budget**.

The project covers the complete analytical workflow from data preparation and feature engineering to exploratory analysis, statistical testing, predictive modelling, and interactive business intelligence. Financial analysis examines revenue, budget, profit, ROI, and success rate. Genre analysis evaluates historical differences in revenue, budget, ROI, and success rate. Audience analysis investigates relationships involving popularity, ratings, vote counts, revenue, and success. Correlation analysis is used to identify relationships among numerical variables, while an independent t-test, Chi-Square test, and Pearson correlation provide statistical evidence for selected research questions.

The predictive scope includes a Random Forest classifier using budget, popularity, runtime, vote average, and primary genre as model inputs. Model performance is assessed using accuracy, precision, recall, F1-score, ROC-AUC, confusion matrix, ROC curve, and feature importance. The final results are delivered through an interactive Streamlit dashboard for portfolio-level analysis and movie-level exploration.

The analysis is intended for historical benchmarking and decision support rather than guaranteed prediction of future box-office performance. Factors not available in the dataset, such as marketing expenditure, cast influence, director influence, distribution strategy, competition, and market conditions, remain outside the current analytical scope.

Table: Scope of the MovieIQ Analysis

Scope Area	Variables / Methods Covered	Analytical Purpose
Data Foundation	Movie attributes, budget, revenue, genre, popularity, ratings, runtime	Establish a reliable analytical dataset
Financial Analysis	Budget, revenue, profit, ROI, success rate	Evaluate financial performance and investment efficiency
Genre Intelligence	Genre, average revenue, average budget, ROI, success rate	Identify historical genre-level performance patterns
Audience Intelligence	Popularity, vote average, vote count, revenue, success	Examine audience-related financial patterns
Success Analysis	Revenue > Budget	Classify movies as successful or unsuccessful
Correlation Analysis	Pearson correlation	Measure linear relationships between numerical variables
Hypothesis Testing	Independent t-test, Chi-Square test	Evaluate statistical differences and associations
Predictive Analytics	Random Forest Classifier	Provide historical success-screening support
Model Evaluation	Accuracy, precision, recall, F1, ROC-AUC, confusion matrix, ROC curve	Assess classification performance
Interactive BI	Streamlit dashboard, filters, movie explorer, predictive interface	Convert analysis into interactive decision support
Out of Scope	Marketing cost, cast/director influence, distribution, competition, market conditions	Variables unavailable in the current dataset

Chapter 2 — Data & Methodology

5. Dataset Description

The MovieIQ project uses a historical movie-level dataset containing **2,000 movie records and 19 available fields**. The dataset provides the core information required to examine financial performance, audience engagement, genre patterns, statistical relationships, and historical movie success. The principal variables include movie title, production budget, revenue, popularity, runtime, vote average, vote count, genres, and release date. These fields provide a foundation for analysing both financial outcomes and audience-related characteristics.

Before analysis, the dataset is processed dynamically within the MovieIQ application. Numeric variables such as budget, revenue, popularity, runtime, vote average, and vote count are converted into appropriate numerical formats, while release date is converted into a date format and used to derive release year where available. Genre information is cleaned and transformed to create a primary genre variable for comparative and predictive analysis.

The dataset is also enriched with analytical variables. Movie success is calculated as **1 when Revenue > Budget and 0 when Revenue ≤ Budget**. Profit is calculated as Revenue minus Budget, while ROI is calculated as (Revenue − Budget) / Budget. Additional analytical segments are created for budget, rating, popularity, and revenue to support grouped comparisons.

The project records **2,000 valid financial records** for the core financial and success analysis. The prepared dataset is subsequently used throughout the exploratory analysis, statistical testing, Random Forest classification, and interactive Streamlit dashboard. This structured preparation ensures that the same analytical definitions and derived variables are consistently applied across the MovieIQ workflow.

6. Data Preparation & Quality

Data preparation is a critical stage of the MovieIQ analytical workflow because the reliability of financial analysis, statistical testing, and predictive modelling depends on the consistency and quality of the underlying data. The project processes the movie dataset dynamically before performing downstream analysis, ensuring that variables are available in suitable formats and that derived analytical fields are created consistently.

The preparation process begins with the conversion of numeric variables such as budget, revenue, popularity, runtime, vote average, and vote count into appropriate numerical formats. Release-date information is converted into a date format, allowing release year to be derived where available. Genre information is cleaned and transformed so that a primary genre can be extracted for comparative analysis and Random Forest modelling.

MovieIQ also creates analytical variables required for the business analysis. The success target is defined as 1 when Revenue > Budget and 0 otherwise. Profit is calculated as

Revenue − Budget, while ROI is calculated as (Revenue − Budget) / Budget. Additional segments are created for budget, rating, popularity, and revenue to support grouped analysis.

For quality validation, the prepared dataset contains **2,000 movie records and 19 fields**, with **2,000 valid financial records** used for the core financial and success analysis. The dashboard's Data Quality section provides checks for dataset dimensions, missing values, duplicate rows, and prepared-data consistency.

The preparation stage therefore establishes a consistent analytical foundation for exploratory analysis, statistical testing, predictive modelling, and interactive dashboard reporting. It also ensures that the same definitions of success, profit, ROI, and genre are applied throughout the MovieIQ platform.

Table: Before and After Data Preparation

Data Preparation Area	Before Preparation	After Preparation
Dataset Structure	Raw movie records and available fields	**2,000 records × 19 fields**
Numeric Variables	May require type conversion	Budget, revenue, popularity, runtime, ratings and vote counts converted to numeric formats
Release Date	Raw date information	Converted to date format; release year derived where available
Genre Data	Raw genre information	Cleaned and transformed into a **primary genre**
Success Target	Not available as analytical target	**Success = 1 if Revenue > Budget; otherwise 0**
Profit	Not directly available as derived measure	**Profit = Revenue − Budget**
ROI	Not directly available as derived measure	**ROI = (Revenue − Budget) / Budget**
Budget Segmentation	Continuous budget values	Budget segments created for comparison
Rating Segmentation	Continuous vote-average values	Rating segments created for grouped analysis
Popularity Segmentation	Continuous popularity values	Popularity segments created for grouped analysis
Revenue Segmentation	Continuous revenue values	Revenue segments created for grouped analysis
Financial Records	Raw financial observations	**2,000 valid financial records** used for core analysis
Quality Checks	Raw quality state	Missing values, duplicate rows, dimensions, and prepared data can be reviewed through the dashboard

7. Feature Engineering

Feature engineering is an important stage of the MovieIQ workflow because the original movie attributes do not directly provide all the measures required for financial analysis, statistical testing, and predictive modelling. The project therefore transforms the prepared dataset into additional analytical features that provide clearer measures of movie performance and create consistent inputs for downstream analysis.

The first major derived variable is **Success**, which serves as the target variable for classification. A movie is assigned a value of 1 when its reported revenue exceeds its production budget and 0 otherwise. This definition enables successful and unsuccessful movies to be compared consistently across the analysis.

Two important financial features are also created. **Profit** is calculated as Revenue − Budget, providing the revenue-minus-budget difference for each movie. **ROI** is calculated as (Revenue − Budget) / Budget, allowing financial efficiency to be compared across movies with different investment levels.

Genre information is transformed to create a **Primary Genre** field, which supports genre-level analysis and is also used as a categorical input in the Random Forest model. The project additionally creates analytical segments for **Budget, Rating, Popularity, and Revenue**, enabling movies to be grouped into meaningful ranges for comparative analysis.

For predictive modelling, the engineered dataset provides the required model features: **budget, popularity, runtime, vote average, and primary genre**. These variables are processed into a suitable modelling structure, including categorical encoding for primary genre. This feature engineering process creates a consistent analytical layer connecting the raw dataset with financial analysis, statistical testing, visualization, and machine learning.

Overall, the feature engineering stage converts raw movie attributes into meaningful business and modelling variables while maintaining the project's defined success criterion and financial measurement framework.

Table: MovieIQ Feature Engineering

Feature	Type	Calculation / Transformation	Purpose
Success	Binary	1 if Revenue > Budget; otherwise 0	Defines the movie success target
Profit	Numerical	Revenue − Budget	Measures revenue-minus-budget outcome
ROI	Numerical	(Revenue − Budget) / Budget	Measures investment efficiency
Primary Genre	Categorical	Extracted from genre information	Supports genre analysis and modelling
Release Year	Numerical	Derived from release date where available	Enables time-based analysis
Budget Segment	Categorical	Budget grouped into analytical ranges	Enables budget-level comparisons
Rating Segment	Categorical	Vote average grouped into ranges	Supports audience-rating comparisons
Popularity Segment	Categorical	Popularity grouped into ranges	Supports audience engagement analysis
Revenue Segment	Categorical	Revenue grouped into ranges	Supports revenue-level comparisons
Encoded Genre Features	Numerical	Primary genre converted using categorical encoding	Makes genre suitable for Random Forest
Model Feature Set	Mixed	Budget, Popularity, Runtime, Vote Average, Primary Genre	Provides inputs for predictive classification

8. Analytical Methodology

The MovieIQ analytical methodology follows a structured end-to-end workflow designed to convert historical movie data into financial, statistical, audience, and predictive insights. The process begins with data preparation and feature engineering, followed by exploratory analysis, financial analysis, statistical testing, predictive modelling, and interactive business intelligence delivery.

First, the prepared dataset is examined using descriptive and exploratory analysis to understand movie characteristics and identify performance patterns. Financial analysis focuses on budget, revenue, profit, ROI, and the defined success outcome. Budget and revenue are compared using visual analysis and Pearson correlation to assess the strength and significance of their linear relationship.

Genre intelligence is developed by comparing movie count, average revenue, average budget, ROI, and success rate across primary genres. Audience intelligence examines popularity, vote average, vote count, revenue, and success to identify potential relationships between audience signals and financial outcomes. A correlation matrix is also used to examine relationships among key numerical variables.

Statistical analysis extends the exploratory findings by applying an independent t-test, Chi-Square test, and Pearson correlation at a significance level of $\alpha = 0.05$. The t-test evaluates whether average popularity differs between successful and unsuccessful movies, while the Chi-Square test examines the association between genre and success. Pearson correlation evaluates the linear relationship between budget and revenue.

For predictive analytics, a Random Forest classifier is trained using budget, popularity, runtime, vote average, and primary genre. The model uses balanced class weights and is evaluated using accuracy, precision, recall, F1-score, ROC-AUC, confusion matrix, ROC curve, and feature importance. Finally, the analytical outputs are integrated into a Streamlit dashboard, enabling users to interact with portfolio-level analysis, movie-level exploration, statistical results, and predictive screening.

The methodology therefore connects descriptive evidence, statistical validation, and predictive modelling while maintaining the distinction between historical association and guaranteed commercial causation.

Table: MovieIQ Analytical Methodology

Method / Stage	Technique / Approach	Purpose	Output
Data Preparation	Type conversion, validation, quality checks	Establish a reliable analytical dataset	Prepared movie dataset
Feature Engineering	Success, Profit, ROI, Primary Genre, analytical segments	Create business and modelling variables	Derived analytical features
Exploratory Analysis	Descriptive statistics and visual analysis	Understand distributions and performance patterns	EDA findings
Financial Analysis	Budget, Revenue, Profit, ROI, Success Rate	Evaluate financial performance and efficiency	Financial insights
Budget–Revenue Analysis	Scatter plot + Pearson correlation	Measure relationship between budget and revenue	Correlation coefficient and p-value
Genre Analysis	Revenue, budget, ROI and success comparisons	Identify historical genre patterns	Genre intelligence
Audience Analysis	Popularity, Vote Average, Vote Count	Examine audience-related performance patterns	Audience insights
Correlation Analysis	Correlation matrix	Identify relationships among numerical variables	Potential success drivers
Independent T-Test	Successful vs unsuccessful popularity	Test difference in group means	t-statistic and p-value
Chi-Square Test	Genre vs Success	Test categorical association	χ^2 statistic and p-value
Random Forest	Classification using selected movie features	Provide predictive screening support	Success/failure predictions
Model Evaluation	Accuracy, Precision, Recall, F1, ROC-AUC	Assess classification quality	Performance metrics
Model Diagnostics	Confusion Matrix, ROC Curve, Feature Importance	Examine model behaviour and important features	Diagnostic insights
Interactive BI	Streamlit dashboard	Deliver analytical results interactively	Decision-support platform

Chapter 3 — Exploratory & Business Analysis

9. Portfolio Financial Performance

The Portfolio Financial Performance analysis evaluates the overall financial characteristics of the 2,000 movies included in the MovieIQ dataset. The analysis focuses on total revenue, total production budget, revenue-minus-budget profit, average return on investment (ROI), and the proportion of movies meeting the project's analytical definition of success. These measures provide a portfolio-level view of financial scale, investment requirements, and historical performance.

The analysed portfolio generated approximately **$356.11 billion in total revenue** against a combined production budget of approximately **$203.84 billion**. The resulting revenue-minus-budget profit was approximately **$152.27 billion**. This indicates that, at the portfolio level, reported revenue exceeded production budget by a substantial amount. However, this measure should not be interpreted as complete accounting profit because the analysis does not include marketing, distribution, taxation, financing, or revenue-sharing costs.

The portfolio recorded an **average ROI of 75.4%**, providing an efficiency measure that complements absolute revenue. ROI is particularly useful because it allows movies with different budget levels to be compared on a relative investment basis. A high-revenue movie is not necessarily the most efficient investment, making ROI an important complementary KPI for project evaluation.

Using the MovieIQ success definition of **Revenue > Budget**, approximately **80.7% of movies were classified as successful**. This indicates that most movies in the analysed historical sample exceeded their production budget in reported revenue. However, this result represents an analytical success measure rather than a definitive statement of commercial profitability.

Overall, the financial analysis establishes a strong baseline for the subsequent budget–revenue relationship, genre intelligence, audience analysis, and predictive modelling. It demonstrates the importance of evaluating both **financial scale and investment efficiency** when assessing historical movie performance.

Table: Portfolio Financial KPIs

Financial KPI	Result	Interpretation
Movies Analysed	**2,000**	Total historical movie records analysed
Total Revenue	**$356.11B**	Overall reported revenue across the portfolio
Total Budget	**$203.84B**	Combined production budget
Total Profit*	**$152.27B**	Revenue minus production budget
Average ROI	**75.4%**	Average return relative to production budget
Success Rate*	**80.7%**	Percentage of movies where Revenue > Budget
Financial Records	**2,000**	Valid financial records used in the analysis

10. Budget–Revenue Relationship

The Budget–Revenue Relationship analysis examines whether movies with higher production budgets tend to generate higher reported revenue. This relationship is important because production budget represents the primary financial investment considered in the MovieIQ dataset, while revenue represents the reported financial outcome. Understanding their relationship can help decision-makers evaluate historical investment patterns and establish appropriate benchmarks for future projects.

MovieIQ analyses the relationship using a **Budget vs Revenue scatter plot** supported by **Pearson correlation analysis**. The visual comparison provides an overview of how revenue changes across different budget levels, while Pearson correlation quantifies the strength and direction of the linear relationship. The analysis produces a correlation coefficient of **r = 0.760** with a **p-value approximately equal to 0.000**, indicating a strong positive and statistically significant linear association between budget and revenue in the analysed sample.

The result suggests that movies with larger production budgets tend to be associated with higher reported revenues. This finding provides evidence that investment scale and revenue are closely related within the historical dataset. However, the relationship should be interpreted carefully. A statistically significant correlation does not establish that increasing a movie's budget will directly cause its revenue to increase. Revenue can also be influenced by factors that are not included in the current dataset, such as marketing expenditure, cast and director influence, distribution strategy, competition, release timing, and broader market conditions.

From a business perspective, the finding supports the use of budget and revenue together when benchmarking historical movie performance. However, budget should not be considered in isolation. Decision-makers should also examine ROI, popularity, ratings, genre characteristics, and other available indicators before evaluating a potential project.

Overall, the analysis identifies budget as an important historical financial indicator while maintaining the distinction between **statistical association and causal impact**.

11. Genre Intelligence

Genre Intelligence evaluates historical movie performance across different primary genres to identify patterns in revenue generation, investment levels, return efficiency, and the defined success outcome. MovieIQ compares genres using movie count, average revenue, average budget, average ROI, and success rate. This multi-dimensional approach is important because a genre with high average revenue is not necessarily the most efficient or the most successful when investment level is considered.

The analysis indicates that **Horror records the highest average revenue at approximately $263.15M**, followed by **Science Fiction at $246.10M** and **Adventure at $235.64M**. These results indicate that several high-performing genres can generate substantial average revenue within the historical sample. However, revenue should be interpreted together with budget and ROI rather than as a standalone measure of financial attractiveness.

When success rate is considered, **Drama records the highest observed success rate at approximately 86.3%**, followed by **Comedy at 85.9%** and **Horror at 84.1%**. This suggests that these genres have historically achieved the MovieIQ success criterion more frequently within the analysed dataset. However, these descriptive differences do not establish that genre itself causes financial success.

The statistical analysis provides an important qualification to the descriptive findings. The Chi-Square test examining the relationship between primary genre and movie success produces a **p-value of 0.99457**, which is substantially above the 0.05 significance threshold. Therefore, the analysis does not provide statistically significant evidence of an association between genre and the defined success outcome.

From a business perspective, genre performance should therefore be used as a **historical benchmarking and portfolio-comparison tool**, rather than as a standalone success predictor. Producers and investors should combine genre information with budget, audience indicators, ROI, market conditions, and other commercial factors when evaluating potential projects.

Table: Genre Performance

Genre	Avg. Revenue	Success Rate	Key Interpretation
Horror	**$263.15M**	**84.1%**	Highest average revenue; strong observed success rate
Science Fiction	**$246.10M**	—	Second-highest average revenue
Adventure	**$235.64M**	—	Strong average revenue performance
Drama	—	**86.3%**	Highest observed success rate
Comedy	—	**85.9%**	Second-highest observed success rate
Other Genres	Varies	Varies	Performance differs across historical genre groups
Chi-Square Test	—	**p = 0.99457**	No statistically significant genre–success association

12. Audience Intelligence

Audience Intelligence examines whether audience-related indicators are associated with movie financial performance and the defined success outcome. MovieIQ focuses on **popularity, vote average, vote count, revenue, and success** to understand how audience engagement and reception vary across historical movie records. These measures provide a complementary perspective to financial analysis because commercial performance may be influenced not only by production investment but also by audience interest and perceived movie quality.

The analysis uses comparative visualizations, including **Popularity vs Revenue** and **Rating vs Revenue**, to examine how audience indicators relate to financial outcomes. Popularity provides an indicator of audience interest, while vote average represents the average audience rating and vote count provides an indication of the volume of audience participation. These variables are analysed alongside revenue and the success classification to identify observable historical patterns.

The statistical analysis provides additional evidence for the popularity comparison. The independent t-test compares average popularity between successful and unsuccessful movies and produces a **p-value of 0.03968**. Since this value is below the significance level of **α = 0.05**, the result provides statistically significant evidence that average popularity differs between the two success groups. However, statistical significance does not establish that popularity directly causes financial success.

Rating and popularity should therefore be interpreted as supporting audience signals rather than standalone predictors. A movie with high audience interest or ratings may perform differently depending on budget, genre, release conditions, marketing, distribution, and other factors that are not captured in the current dataset.

From a business perspective, audience intelligence can support historical benchmarking and help producers, studios, and marketing teams understand audience-related patterns. Combining audience indicators with financial metrics, genre characteristics, and predictive outputs provides a more comprehensive basis for project screening and decision support.

Table: Audience Metrics

Audience Metric	Analytical Role	Business Interpretation
Popularity	Measures audience interest/engagement	Higher or lower popularity can be compared across success groups
Vote Average	Measures average audience rating	Provides an indicator of perceived audience reception
Vote Count	Measures volume of audience participation	Provides context for the strength of rating-based evidence
Revenue	Financial outcome	Used to examine whether audience indicators relate to commercial performance
Success	Binary outcome: Revenue > Budget	Enables comparison of audience characteristics between successful and unsuccessful movies
Popularity T-Test	Independent t-test	**p = 0.03968**, indicating a statistically significant difference between success groups
Popularity vs Revenue	Comparative visualization	Examines the relationship between audience interest and financial performance
Rating vs Revenue	Comparative visualization	Examines whether audience ratings show observable financial patterns

13. Success Driver Analysis

The Success Driver Analysis examines the characteristics that show meaningful relationships with movie performance and the defined success outcome. MovieIQ evaluates financial, audience, and movie-level variables including budget, revenue, profit, ROI, popularity, runtime, vote average, vote count, and success. A correlation matrix is used to identify the direction and strength of relationships among numerical variables, while comparisons

between successful and unsuccessful movies provide additional context for interpreting potential success drivers.

The analysis indicates that **revenue, profit, and ROI** have strong relationships with the defined success outcome. This result is expected to some extent because success is directly constructed from the relationship between revenue and budget. Therefore, these variables should be treated primarily as **performance indicators rather than independent predictive drivers**. Using them as predictors of the same success definition could introduce target leakage in a predictive model.

Among the audience-related variables, **popularity** provides meaningful analytical evidence. The independent t-test reports a **p-value of 0.03968**, indicating a statistically significant difference in average popularity between successful and unsuccessful movies at $\alpha = 0.05$. This suggests that audience interest is relevant to historical success-group differences, although it does not establish a causal relationship.

Budget also emerges as an important financial indicator because it has a strong positive relationship with revenue, with Pearson correlation of $\mathbf{r = 0.760}$ and $\mathbf{p \approx 0.000}$. However, budget should not be interpreted as a direct cause of success. Runtime and vote average provide additional contextual information, but their relationships should be interpreted cautiously rather than treated as standalone success determinants.

Overall, MovieIQ identifies potential success-related indicators rather than definitive causal drivers. The findings support a multi-factor approach combining financial scale, audience signals, genre context, and other commercial variables. Future modelling should incorporate additional business variables to improve the ability to distinguish genuine predictive factors from variables that are mathematically related to the success definition.

Table: Success Driver Ranking

Rank	Variable / Indicator	Analytical Evidence	Driver Interpretation
1	**Revenue**	Directly linked to success definition	Strong performance indicator; not an independent predictive driver
2	**Profit**	Derived from Revenue − Budget	Strong outcome indicator; affected by target definition
3	**ROI**	Derived from Revenue and Budget	Strong investment-efficiency indicator; potential target leakage
4	**Budget**	Pearson **r = 0.760** with Revenue; $p \approx 0.000$	Strong financial indicator, but association does not prove causation
5	**Popularity**	T-test **p = 0.03968**	Statistically significant difference between success groups
6	**Vote Average**	Audience reception indicator	Useful contextual signal; not sufficient as a standalone predictor
7	**Vote Count**	Measures audience participation	Provides supporting audience context
8	**Runtime**	Movie characteristic	Contextual variable with limited standalone interpretation

Chapter 4 — Statistical Analysis

14. Hypothesis Testing

Hypothesis testing is used in MovieIQ to move beyond descriptive visual patterns and determine whether selected relationships or group differences have sufficient statistical evidence. The analysis applies three statistical techniques: an independent t-test, a Chi-Square test, and Pearson correlation. A common significance level of $\alpha = 0.05$ is used throughout the testing framework. A p-value below 0.05 is interpreted as statistically significant evidence against the corresponding null hypothesis, while a p-value above 0.05 indicates insufficient evidence to reject the null hypothesis.

The **independent t-test** examines whether average popularity differs between successful and unsuccessful movies. The null hypothesis states that the mean popularity of the two groups is equal, while the alternative hypothesis states that the means differ. The analysis produces a **p-value of 0.03968**, which is below 0.05. Therefore, the null hypothesis is rejected, providing statistically significant evidence of a difference in average popularity between the two groups.

The **Chi-Square test** evaluates whether primary genre and movie success are statistically associated. The null hypothesis states that genre and success are independent. The test produces a **p-value of 0.99457**, which is substantially greater than 0.05. Therefore, the null hypothesis is not rejected, indicating insufficient statistical evidence of an association between genre and the defined success outcome in the analysed sample.

The **Pearson correlation test** evaluates the linear relationship between budget and revenue. The analysis reports **r = 0.760** with **p ≈ 0.000**, indicating a strong positive and statistically significant linear association. However, none of these tests establishes causation. The results should therefore be interpreted as statistical evidence of differences or associations within the historical dataset rather than proof that one variable directly causes another.

Table: Hypothesis Testing Results

Test	Research Question	Null Hypothesis (H_0)	Result	Decision	Interpretation
Independent T-Test	Does popularity differ between successful and unsuccessful movies?	Mean popularity is equal between the two groups	**p = 0.03968**	**Reject H_0**	Significant difference in average popularity
Chi-Square Test	Is primary genre associated with movie success?	Genre and success are independent	**p = 0.99457**	**Fail to Reject H_0**	No significant genre–success association
Pearson Correlation	Is budget linearly related to revenue?	No linear relationship exists between budget and revenue	**r = 0.760, p ≈ 0.000**	**Reject H_0**	Strong, statistically significant positive relationship

15· Statistical Findings & Interpretation

The statistical analysis provides evidence-based interpretation of the relationships identified during the exploratory and business analysis stages. MovieIQ applies an independent t-test, Chi-Square test, and Pearson correlation using a significance level of **α = 0.05**. These tests help determine whether selected differences and relationships are statistically supported within the analysed historical movie dataset.

The independent t-test evaluates whether average popularity differs between successful and unsuccessful movies. The test produces a **p-value of 0.03968**, which is below the 0.05 significance threshold. Therefore, the null hypothesis is rejected, providing statistically significant evidence that the two success groups have different average popularity levels. However, the result indicates a difference between groups and does not demonstrate that popularity directly causes movie success.

The Chi-Square test examines the relationship between primary genre and the defined success outcome. The resulting **p-value of 0.99457** is substantially higher than 0.05. Consequently, the null hypothesis is not rejected, meaning the analysis does not provide sufficient statistical evidence of an association between genre and movie success. Descriptive differences between genres should therefore be interpreted as historical patterns rather than statistically established determinants of success.

Pearson correlation provides the strongest statistical relationship identified in the analysis. The correlation between budget and revenue is **r = 0.760**, with **p ≈ 0.000**, indicating a strong positive and statistically significant linear association. This suggests that higher-budget movies tend to have higher reported revenue within the dataset. However, correlation does not establish causation, and other commercial factors may influence revenue.

Overall, the statistical findings strengthen the MovieIQ analysis by distinguishing statistically supported relationships from descriptive patterns. The results should be used as decision-support evidence rather than as proof of causal relationships or guaranteed future movie performance.

Table: Statistical Findings

Statistical Test	Key Result	Statistical Decision	Business Interpretation
Independent T-Test	**p = 0.03968**	Reject H_0	Average popularity differs significantly between successful and unsuccessful movies
Chi-Square Test	**p = 0.99457**	Fail to Reject H_0	No statistically significant association between primary genre and movie success
Pearson Correlation	**r = 0.760; p ≈ 0.000**	Reject H_0	Strong, statistically significant positive relationship between budget and revenue
Significance Level	**α = 0.05**	—	Common threshold used for interpreting statistical evidence
Overall Finding	—	—	Statistical evidence supports selected relationships but does not establish causation

Chapter 5 — Predictive Analytics

16· Random Forest Model

MovieIQ uses a **Random Forest Classifier** to provide predictive screening support by classifying movies into the defined success and failure categories. The target variable is based on the project's analytical rule, where a movie is classified as successful when **Revenue > Budget** and unsuccessful otherwise. The model is designed to identify historical patterns in movie characteristics that may help distinguish between these two outcomes.

The model uses five input features: **budget, popularity, runtime, vote average, and primary genre**. These variables represent financial investment, audience interest, movie characteristics, audience reception, and genre context. Primary genre is treated as a categorical variable and is transformed into numerical dummy variables before being supplied to the classifier.

MovieIQ implements the Random Forest model directly within the Streamlit application, meaning that the model is trained dynamically rather than relying on a separately stored model file. The configuration uses **300 decision trees**, a maximum tree depth of **10**, minimum samples split of **5**, and minimum samples leaf of **2**. The model also uses **balanced class weights** to help address potential differences in the distribution of success and failure classes. A **random state of 42** is used to support reproducibility, while parallel processing is enabled to improve computational efficiency.

The model is trained using a stratified train-test split so that the relative distribution of the target classes is maintained across the training and testing datasets. This supports a more consistent evaluation of classification performance.

The Random Forest is intended as a **decision-support and historical screening mechanism**, not as a guaranteed box-office forecasting system. Its predictions should be interpreted alongside financial metrics, audience indicators, genre context, and commercial knowledge. The model's limitations are particularly important because several real-world factors, including marketing expenditure, cast influence, distribution strategy, competition, and market conditions, are not represented in the current dataset.

Table: Random Forest Model Configuration

Configuration	Value	Purpose
Algorithm	Random Forest Classifier	Predicts movie success/failure
Target Variable	Success	1 if Revenue > Budget; otherwise 0
Model Features	Budget, Popularity, Runtime, Vote Average, Primary Genre	Historical movie characteristics used for classification
Categorical Encoding	One-hot / dummy encoding	Converts Primary Genre into numerical features
Number of Estimators	**300**	Number of decision trees
Maximum Depth	**10**	Controls tree complexity

Minimum Samples Split	**5**	Minimum observations required to split a node
Minimum Samples Leaf	**2**	Minimum observations required in a leaf
Class Weight	**Balanced**	Helps address class imbalance
Random State	**42**	Supports reproducible results
Training Method	Stratified train-test split	Maintains class proportions during evaluation
Parallel Processing	Enabled (`n_jobs=-1`)	Improves training efficiency
Training Approach	Dynamic in Streamlit	Model is trained within the application

17· Model Evaluation

The Random Forest model developed in MovieIQ is evaluated using multiple performance metrics to provide a balanced assessment of its classification capability. Instead of relying only on accuracy, the evaluation considers **precision, recall, F1-score, and ROC-AUC**, supported by a confusion matrix and ROC curve. This approach is important because a single metric may not fully represent how well the model distinguishes successful and unsuccessful movies.

The model reports an **accuracy of 65.0%**, indicating that approximately 65% of the evaluated observations were classified correctly. Precision is **80.2%**, showing that a relatively high proportion of movies predicted as successful were correctly identified. Recall is **75.2%**, indicating that the model identified a substantial proportion of the actual successful movies. The resulting **F1-score of 77.6%** provides a combined measure of precision and recall and indicates moderate classification performance based on the displayed evaluation.

The confusion matrix provides a more detailed view of correct and incorrect classifications by comparing actual outcomes with predicted outcomes. It helps identify whether the model performs differently when identifying successful and unsuccessful movies. The ROC curve provides another perspective by examining the model's discrimination behaviour across classification thresholds.

However, the reported **ROC-AUC of 0.473** indicates limited discrimination capability. Since the value is below 0.5, the model's ability to rank successful movies above unsuccessful movies is weak in the evaluated results. Therefore, the relatively stronger precision, recall, and F1-score should not be interpreted as evidence of a highly reliable forecasting model.

Overall, the Random Forest provides useful exploratory screening capability but should be treated as a **decision-support tool rather than a guaranteed movie-success predictor**. Its outputs should be combined with financial analysis, audience indicators, genre context, and additional commercial information before making real-world investment decisions.

Table: Evaluation Metrics

Evaluation Metric	Result	Interpretation
Accuracy	**65.0%**	Percentage of overall predictions classified correctly
Precision	**80.2%**	Proportion of predicted successful movies that were actually successful
Recall	**75.2%**	Proportion of actual successful movies correctly identified
F1-Score	**77.6%**	Combined measure of precision and recall
ROC-AUC	**0.473**	Indicates limited discrimination/ranking capability
Confusion Matrix	Included	Shows correct and incorrect predictions by class
ROC Curve	Included	Shows model discrimination across classification thresholds

18. Feature Importance & Prediction Interpretation

Feature importance analysis is used in MovieIQ to understand which input variables contribute most to the Random Forest classification decisions. The model uses **budget, popularity, runtime, vote average, and primary genre** as predictive features. Unlike correlation analysis, feature importance reflects the contribution of variables within the trained Random Forest model and should therefore be interpreted as a model-specific measure rather than proof of causation.

Among the selected variables, **budget and popularity** are particularly important from a business perspective because they represent financial investment and audience interest respectively. Budget is also strongly associated with revenue in the broader statistical analysis, while popularity shows a statistically significant difference between successful and unsuccessful movies according to the independent t-test. These findings provide useful context when interpreting the model's predictive behaviour.

Runtime and vote average provide additional movie-level information. Runtime represents the structural characteristics of a movie, while vote average reflects audience reception. Their importance in the model indicates whether these characteristics contribute to classification decisions within the trained dataset. However, a lower feature-importance score does not necessarily mean that a variable has no business value; it only indicates a smaller contribution relative to the other features under this particular model configuration.

Primary genre is treated as a categorical feature and encoded before modelling. Although genre may show descriptive differences in revenue and success rates, the Chi-Square test found **no statistically significant association between genre and the defined success outcome (p = 0.99457)**. Therefore, genre should be interpreted as contextual information rather than an independent determinant of success.

Prediction interpretation should also consider the model's overall performance. With an **ROC-AUC of 0.473**, the Random Forest demonstrates limited discrimination capability. Consequently, feature importance is best used to understand model behaviour and generate analytical insights rather than to claim that any feature guarantees movie success.

Table: Feature Importance Ranking

Rank	Feature	Business Role	Prediction Interpretation
1	**Budget**	Financial investment	Indicates the contribution of production investment to model decisions
2	**Popularity**	Audience interest	Represents audience engagement and historical success-group differences
3	**Vote Average**	Audience reception	Captures perceived movie quality through ratings
4	**Runtime**	Movie characteristic	Provides contextual information about movie structure
5	**Primary Genre**	Content category	Provides genre context after categorical encoding

Chapter 6 — Business Intelligence

19. Streamlit Dashboard

The MovieIQ Streamlit Dashboard serves as the interactive business intelligence layer of the project, transforming the analytical results into an accessible and user-friendly decision-support application. The dashboard integrates financial analysis, audience intelligence, genre analysis, statistical testing, predictive modelling, and movie-level exploration within a single interface. Users can navigate between analytical modules and interact with visualizations to examine historical movie performance from different perspectives.

The **Executive Overview** provides a high-level summary of the movie portfolio through key performance indicators such as total movies, revenue, budget, ROI, and success rate. The **Financial Analysis** module focuses on budget, revenue, profit, ROI, and financial relationships. **Genre Intelligence** enables comparison of movie performance across primary genres, while **Audience Intelligence** examines popularity, ratings, vote counts, and revenue patterns.

The **Statistical Analysis** module presents hypothesis-testing results, including the independent t-test, Chi-Square test, and Pearson correlation, allowing users to distinguish statistically supported relationships from descriptive observations. The **Predictive Analytics** module presents the Random Forest model, its evaluation metrics, feature importance, and prediction-related outputs. The **Movie Explorer** provides a more detailed movie-level view, enabling users to investigate individual records and their characteristics.

The dashboard also includes data quality and preparation information to improve transparency around the analytical dataset. Interactive charts, KPI cards, filters, and structured sections make the analytical findings easier to interpret and communicate.

Overall, the Streamlit application connects the complete MovieIQ workflow—from **data to analysis, statistical evidence, prediction, and business insight**—within one interactive platform. It is designed primarily as a decision-support and analytical exploration tool rather than a system that guarantees future movie success.

Table: Streamlit Dashboard Modules

Dashboard Module	Main Content	Business Purpose
Executive Overview	KPIs, portfolio summary, key trends	Provides a high-level performance snapshot
Financial Analysis	Budget, revenue, profit, ROI	Evaluates financial performance and investment efficiency
Genre Intelligence	Genre performance, revenue, ROI, success rate	Compares historical genre patterns
Audience Intelligence	Popularity, ratings, vote count, revenue	Analyses audience-related performance indicators

Statistical Analysis	T-Test, Chi-Square, Pearson correlation	Validates selected relationships statistically
Predictive Analytics	Random Forest, predictions, model metrics	Provides historical success-screening support
Feature Importance	Model feature ranking	Explains variables contributing to model decisions
Movie Explorer	Individual movie details and metrics	Enables movie-level investigation
Data Quality / Explorer	Dataset dimensions, missing values, duplicates, data preview	Improves transparency and data understanding
Interactive Controls	Filters, selectors, charts, KPI cards	Enables user-driven exploration of MovieIQ insights

20. Business Insights & Recommendations

The MovieIQ analysis provides several business insights that can support more structured decision-making in film investment and project screening. The portfolio generated **$356.11 billion in revenue** against **$203.84 billion in budget**, with an average ROI of **75.4%** and an analytical success rate of **80.7%**. These results indicate strong historical portfolio performance, although revenue-minus-budget should not be treated as complete accounting profit because marketing, distribution, taxes, and other commercial costs are not included.

The **budget–revenue relationship** is one of the strongest findings, with Pearson correlation of **r = 0.760 (p ≈ 0.000)**. This suggests that higher-budget movies tend to generate higher reported revenue, but it should not be interpreted as proof that increasing budget directly causes higher revenue. Investment decisions should therefore balance budget scale with expected ROI and other available indicators.

Audience intelligence also provides useful evidence. The independent t-test produced **p = 0.03968**, indicating a statistically significant difference in popularity between successful and unsuccessful movies. This suggests that audience interest can be considered as one supporting indicator during project evaluation. In contrast, the Chi-Square test produced **p = 0.99457**, indicating no statistically significant association between primary genre and the defined success outcome.

The Random Forest model provides additional screening support, but its **65.0% accuracy and 0.473 ROC-AUC** indicate limited predictive discrimination. Therefore, model predictions should not be used as standalone investment decisions.

Table: Business Recommendations

Insight	Evidence	Recommendation	Business Priority
Budget is strongly related to revenue	r = 0.760, p ≈ 0.000	Evaluate budget alongside expected revenue and ROI rather than increasing budget automatically	**High**
Audience popularity differs by success group	T-test p = 0.03968	Use popularity as a supporting audience-demand indicator	**High**
Genre is not statistically associated with success	Chi-Square p = 0.99457	Avoid selecting projects solely because of genre; evaluate additional factors	**High**
Portfolio shows strong historical performance	80.7% success rate; 75.4% average ROI	Use historical benchmarks when screening and comparing projects	**Medium**
Predictive model has limitations	Accuracy 65.0%; ROC-AUC 0.473	Use Random Forest as a screening aid, not a guaranteed forecasting tool	**High**
Financial efficiency matters	ROI = 75.4%	Compare potential projects using both revenue scale and investment efficiency	**High**
Additional commercial variables are needed	Current dataset limitations	Incorporate marketing, cast, distribution, competition, and release factors in future models	**Medium**

21. Limitations

The MovieIQ project provides a structured framework for analysing historical movie performance; however, several limitations should be considered when interpreting its findings. First, the analysis is based on the variables available in the dataset. Important commercial factors such as **marketing expenditure, distribution costs, cast popularity, director influence, production quality, competition, release timing, and market conditions** are not included. Consequently, the observed relationships may not capture all factors that influence actual movie performance.

Second, MovieIQ defines movie success as **Revenue > Budget**. Although this provides a clear and consistent analytical criterion, it does not represent complete commercial profitability. A movie may generate revenue above its production budget while still making a loss after accounting for marketing, distribution, taxation, financing, theatre revenue sharing, and other costs. Therefore, the reported success rate should be interpreted as an analytical financial indicator rather than a definitive measure of profitability.

Third, several relationships identified in the analysis are **associations rather than causal relationships**. For example, the strong positive relationship between budget and revenue does not prove that increasing production expenditure will automatically increase revenue. Similarly, the significant difference in popularity between success groups does not establish that popularity directly causes success.

The predictive Random Forest model also has limitations. Its reported **65.0% accuracy and 0.473 ROC-AUC** indicate that its classification capability is limited, particularly in terms of distinguishing successful and unsuccessful movies. Therefore, its predictions should be treated as exploratory screening outputs rather than guaranteed forecasts.

Finally, feature importance reflects the behaviour of the trained model and should not be interpreted as causal importance. The project should therefore be viewed primarily as a **historical analytics and decision-support system**, with future improvements requiring richer commercial data, larger and more representative datasets, stronger validation approaches, and additional predictive variables.

22. Future Scope

The future scope of MovieIQ — Film Success Intelligence focuses on improving the depth, accuracy, and practical value of the platform by incorporating richer data, advanced analytical techniques, and more comprehensive predictive capabilities. The current project establishes a foundation using historical movie characteristics, financial indicators, audience metrics, statistical testing, Random Forest classification, and interactive Streamlit visualization. Future development can extend these capabilities to address the limitations identified in the present analysis.

A major opportunity is to **expand the dataset** by incorporating additional commercial and industry variables. Future versions could include marketing and advertising expenditure, distribution costs, opening-weekend performance, cast and director information, production companies, release timing, competition, streaming performance, and regional market information. These variables could provide a more complete representation of the factors influencing commercial movie performance.

The definition of success can also be enhanced. Instead of relying only on **Revenue > Budget**, future versions could incorporate complete profitability measures where reliable cost and revenue-sharing information is available. This would provide a more realistic assessment of financial performance and investment return.

From a predictive analytics perspective, future versions could compare multiple algorithms such as **Logistic Regression, Gradient Boosting, XGBoost, and other ensemble approaches** with appropriate cross-validation and hyperparameter tuning. Model calibration and explainability techniques could also be introduced to improve the reliability and interpretation of predictions.

The Streamlit dashboard could further evolve into a real-time decision-support platform with scenario analysis, budget simulation, probability-based success scoring, and automated recommendation features. Future models could also incorporate current market and audience signals to support more timely project evaluation.

Overall, the future direction of MovieIQ is to evolve from a historical analytics platform into a **more comprehensive, explainable, and data-driven film decision-support system**, while clearly communicating model uncertainty and avoiding unsupported guarantees of movie success.

23· Conclusion

MovieIQ — Film Success Intelligence demonstrates how data analytics, statistical analysis, predictive modelling, and interactive business intelligence can be combined to evaluate historical movie performance and support evidence-based decision-making. The project analysed **2,000 movies** using financial, audience, genre, and movie-level characteristics and transformed the available data into meaningful analytical measures such as revenue, budget, profit, ROI, and the defined success indicator.

The financial analysis showed that the portfolio generated approximately **$356.11 billion in total revenue** against **$203.84 billion in total budget**, with an average ROI of **75.4%** and an analytical success rate of **80.7%**. The Budget–Revenue analysis identified a strong positive relationship between budget and revenue (**r = 0.760, p ≈ 0.000**), demonstrating that investment scale is strongly associated with reported revenue within the historical dataset.

Audience analysis provided additional evidence that popularity differs between successful and unsuccessful movies, supported by an independent t-test with **p = 0.03968**. In contrast, the Chi-Square test produced **p = 0.99457**, indicating no statistically significant association between primary genre and the defined success outcome. These findings demonstrate the importance of distinguishing descriptive patterns from statistically supported relationships.

The Random Forest model provided an additional predictive perspective using budget, popularity, runtime, vote average, and primary genre. Although the model achieved **65.0% accuracy, 80.2% precision, 75.2% recall, and 77.6% F1-score**, its **ROC-AUC of 0.473** indicates limited discrimination capability. Therefore, the model should be used as a screening and decision-support tool rather than a guaranteed predictor of future movie success.

Overall, MovieIQ successfully converts historical movie data into structured financial, audience, statistical, and predictive intelligence. The platform provides a practical foundation for portfolio benchmarking and project evaluation while clearly acknowledging the uncertainty and limitations inherent in movie-success prediction.